

Introduction to Meta-Analysis

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Import libraries

```
library(dplyr)
library(meta)
library(readxl)
```

Load data

```
AcuteManiaP <- AcuteManiaP_raw %>% filter(treat2 == "Risperidone")

str(AcuteManiaP)

## tibble [5 x 15] (S3: tbl_df/tbl/data.frame)
##  $ studlab: num [1:5] 23 24 25 56 58
##  $ treat1 : chr [1:5] "Placebo" "Placebo" "Placebo" "Placebo" ...
##  $ treat2 : chr [1:5] "Risperidone" "Risperidone" "Risperidone" "Risperidone" ...
##  $ ncont1 : num [1:5] 142 72 119 47 138
##  $ mean1  : num [1:5] -10.5 -10.3 -4.8 -8.2 22.1
##  $ sd1    : num [1:5] 15.5 11.9 9.5 10 10.2
##  $ ncont2 : num [1:5] 144 68 127 51 153
##  $ mean2  : num [1:5] -22.7 -14.5 -10.6 -14.3 17
##  $ sd2    : num [1:5] 13.2 12.4 9.5 10 10.2
##  $ event1 : num [1:5] 51 30 29 NA 39
##  $ n1     : num [1:5] 145 76 125 51 140
##  $ event2 : num [1:5] 105 40 55 NA 65
##  $ n2     : num [1:5] 146 75 134 52 154
##  $ AE1    : num [1:5] 5 1 4 0 5
##  $ AE2    : num [1:5] 3 0 0 0 2

nrow(AcuteManiaP)

## [1] 5

pooledSMD1 <- metacont(mean.c = mean1, sd.c = sd1, n.c = ncont1,
  mean.e = mean2, sd.e = sd2, n.e = ncont2,
  data = AcuteManiaP, studlab = studlab, sm = "SMD")

summary(pooledSMD1)

##           SMD           95%-CI %W(common) %W(random)
## 23 -0.8460 [-1.0881; -0.6039]      26.0      23.8
## 24 -0.3446 [-0.6786; -0.0107]      13.6      16.5
```

```
## 25 -0.6086 [-0.8645; -0.3528]      23.3      22.5
## 56 -0.6036 [-1.0091; -0.1980]       9.3      12.6
## 58 -0.5003 [-0.7340; -0.2666]      27.9      24.6
##
## Number of studies: k = 5
## Number of observations: o = 1061 (o.e = 543, o.c = 518)
##
##              SMD              95%-CI      z  p-value
## Common effect model -0.6036 [-0.7270; -0.4802] -9.59 < 0.0001
## Random effects model -0.5943 [-0.7627; -0.4259] -6.92 < 0.0001
##
## Quantifying heterogeneity (with 95%-CIs):
## tau^2 = 0.0158 [0.0000; 0.2555]; tau = 0.1256 [0.0000; 0.5054]
## I^2 = 42.1% [0.0%; 78.7%]; H = 1.31 [1.00; 2.17]
##
## Test of heterogeneity:
##      Q d.f. p-value
## 6.91   4  0.1405
##
## Details of meta-analysis methods:
## - Inverse variance method
## - Restricted maximum-likelihood estimator for tau^2
## - Q-Profile method for confidence interval of tau^2 and tau
## - Calculation of I^2 based on Q
## - Hedges' g (bias corrected standardised mean difference; using exact formulae)
```

5 studies were involved in the meta-analysis the fixed effect model gave an estimate of -0.60 [-0.73; -0.48]
random effects model: -0.59 [-0.76; -0.43]

Both significant

The test of heterogeneity does not highlight evidence of heterogeneity. This doesn't mean that there's no heterogeneity, it might be low power

meta analysis: in this case I specify the estimator PM of tau (also the default) and I ask not to compute a fixed-effect meta-analysis

```
summary(update(pooledSMD1, method.tau = "PM"))
```

```
##              SMD              95%-CI %W(common) %W(random)
## 23 -0.8460 [-1.0881; -0.6039]      26.0      24.0
## 24 -0.3446 [-0.6786; -0.0107]      13.6      16.2
## 25 -0.6086 [-0.8645; -0.3528]      23.3      22.6
## 56 -0.6036 [-1.0091; -0.1980]       9.3      12.2
## 58 -0.5003 [-0.7340; -0.2666]      27.9      24.9
##
## Number of studies: k = 5
## Number of observations: o = 1061 (o.e = 543, o.c = 518)
##
##              SMD              95%-CI      z  p-value
## Common effect model -0.6036 [-0.7270; -0.4802] -9.59 < 0.0001
## Random effects model -0.5952 [-0.7579; -0.4325] -7.17 < 0.0001
##
## Quantifying heterogeneity (with 95%-CIs):
## tau^2 = 0.0134 [0.0000; 0.2555]; tau = 0.1160 [0.0000; 0.5054]
## I^2 = 42.1% [0.0%; 78.7%]; H = 1.31 [1.00; 2.17]
##
```

```
## Test of heterogeneity:
##      Q d.f. p-value
## 6.91    4  0.1405
##
## Details of meta-analysis methods:
## - Inverse variance method
## - Paule-Mandel estimator for tau^2
## - Q-Profile method for confidence interval of tau^2 and tau
## - Calculation of I^2 based on Q
## - Hedges' g (bias corrected standardised mean difference; using exact formulae)

random effects model: -0.60 [-0.76; -0.43] It is very similar to REML case (default) However, with only 5
studies, it's difficult to estimate the heterogeneity (we need at least >5 studies).
```

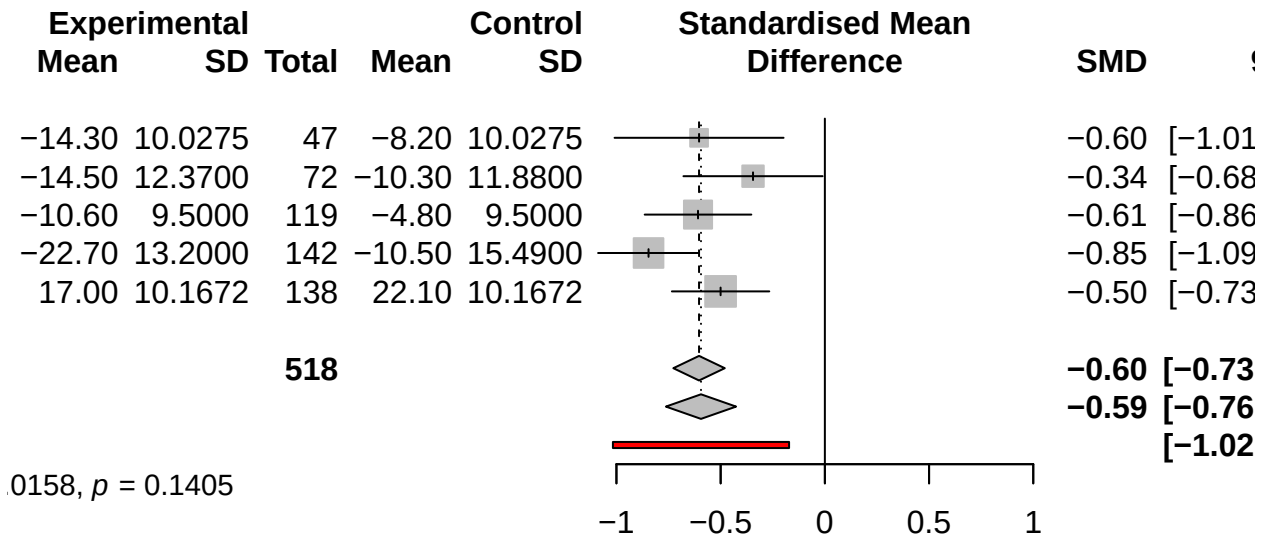
```
summary(update(pooledSMD1, method.random.ci = T))
```

```
##           SMD           95%-CI %W(common) %W(random)
## 23 -0.8460 [-1.0881; -0.6039]      26.0      23.8
## 24 -0.3446 [-0.6786; -0.0107]      13.6      16.5
## 25 -0.6086 [-0.8645; -0.3528]      23.3      22.5
## 56 -0.6036 [-1.0091; -0.1980]       9.3      12.6
## 58 -0.5003 [-0.7340; -0.2666]      27.9      24.6
##
## Number of studies: k = 5
## Number of observations: o = 1061 (o.e = 543, o.c = 518)
##
##           SMD           95%-CI    z|t    p-value
## Common effect model -0.6036 [-0.7270; -0.4802] -9.59 < 0.0001
## Random effects model -0.5943 [-0.8249; -0.3638] -7.16    0.0020
##
## Quantifying heterogeneity (with 95%-CIs):
## tau^2 = 0.0158 [0.0000; 0.2555]; tau = 0.1256 [0.0000; 0.5054]
## I^2 = 42.1% [0.0%; 78.7%]; H = 1.31 [1.00; 2.17]
##
## Test of heterogeneity:
##      Q d.f. p-value
## 6.91    4  0.1405
##
## Details of meta-analysis methods:
## - Inverse variance method
## - Restricted maximum-likelihood estimator for tau^2
## - Q-Profile method for confidence interval of tau^2 and tau
## - Calculation of I^2 based on Q
## - Hartung-Knapp adjustment for random effects model (df = 4)
## - Hedges' g (bias corrected standardised mean difference; using exact formulae)
```

Hartung-Knapp adjustment for random effects model (df = 4)

FOREST PLOTS

```
forest(pooledSMD1, sortvar = ncont1 + ncont2,
       prediction = T, common = T)
```



Meta-analysis with binary outcome

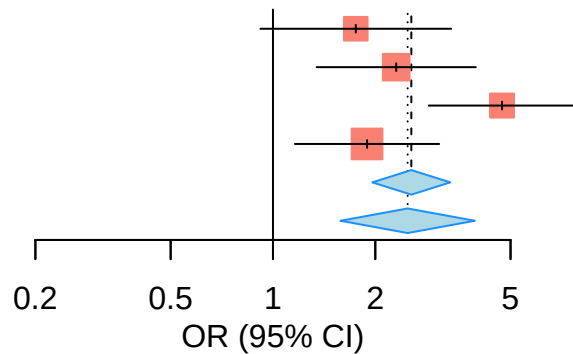
Odds-Ratio as Effect Size

```
pooledOR <- metabin(event.e = event2,
  n.e = n2,
  event.c = event1,
  n.c = n1,
  data = AcuteManiaP,
  studlab = studlab, sm = "OR")
summary(pooledOR)
```

```
##          OR          95%-CI %W(common) %W(random)
## 23 4.7202 [2.8731; 7.7548]    20.7      26.5
## 24 1.7524 [0.9185; 3.3434]    20.0      21.7
## 25 2.3047 [1.3439; 3.9524]    25.4      25.1
## 56    NA
## 58 1.8914 [1.1603; 3.0832]    33.9      26.7
##
## Number of studies: k = 4
## Number of observations: o = 1098 (o.e = 561, o.c = 537)
## Number of events: e = 414
##
##          OR          95%-CI      z  p-value
## Common effect model 2.5529 [1.9603; 3.3248] 6.95 < 0.0001
## Random effects model 2.4903 [1.5804; 3.9242] 3.93 < 0.0001
```

```
##
## Quantifying heterogeneity (with 95%-CIs):
## tau^2 = 0.1391 [0.0000; 2.7687]; tau = 0.3730 [0.0000; 1.6640]
## I^2 = 65.8% [0.0%; 88.4%]; H = 1.71 [1.00; 2.93]
##
## Test of heterogeneity:
##      Q d.f. p-value
## 8.78    3  0.0324
##
## Details of meta-analysis methods:
## - Mantel-Haenszel method (common effect model)
## - Inverse variance method (random effects model)
## - Restricted maximum-likelihood estimator for tau^2
## - Q-Profile method for confidence interval of tau^2 and tau
## - Calculation of I^2 based on Q
forest(pooledOR, sortvar = n1 + n2, layout = "JAMA", allstudies = F,
      fs.study = 10, ff.study = "italic",
      col.diamond = "indianred", col.diamond.lines = "dodgerblue",
      col.square = "salmon", col.square.lines = "salmon")
```

Source	OR (95% CI)
24	1.75 [0.92; 3.34]
25	2.30 [1.34; 3.95]
23	4.72 [2.87; 7.75]
58	1.89 [1.16; 3.08]
Total (common effect)	2.55 [1.96; 3.32]
Total (random effect)	2.49 [1.58; 3.92]



Heterogeneity: $\chi^2_3 = 8.78$ ($P = .03$), $I^2 = 65.8\%$

```
summary(update(pooledOR, sm = "RR"))
```

```
##      RR      95%-CI %W(common) %W(random)
## 23 2.0447 [1.6034; 2.6075]      33.7      34.1
## 24 1.3511 [0.9524; 1.9168]      19.6      21.9
## 25 1.7692 [1.2122; 2.5820]      19.8      19.6
```

```

## 56      NA                0.0      0.0
## 58 1.5152 [1.0954; 2.0957]    26.9    24.3
##
## Number of studies: k = 4
## Number of observations: o = 1098 (o.e = 561, o.c = 537)
## Number of events: e = 414
##
##              RR          95%-CI    z  p-value
## Common effect model  1.7116 [1.4648; 2.0001] 6.76 < 0.0001
## Random effects model 1.6874 [1.3858; 2.0546] 5.21 < 0.0001
##
## Quantifying heterogeneity (with 95%-CIs):
## tau^2 = 0.0142 [0.0000; 0.4305]; tau = 0.1191 [0.0000; 0.6562]
## I^2 = 31.5% [0.0%; 75.4%]; H = 1.21 [1.00; 2.02]
##
## Test of heterogeneity:
##      Q d.f. p-value
## 4.38   3  0.2231
##
## Details of meta-analysis methods:
## - Mantel-Haenszel method (common effect model)
## - Inverse variance method (random effects model)
## - Restricted maximum-likelihood estimator for tau^2
## - Q-Profile method for confidence interval of tau^2 and tau
## - Calculation of I^2 based on Q

```